A cut and independent-set bound for sparse halves

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Abstract

We derive the partial bound $2587/100000 = 0.02587$ for sparse halves of every finite triangle-free graph, assuming the Balogh–Clemen–Lidický max-cut theorem. The argument is elementary: no flag algebra or semidefinite certificate is used. Its one new ingredient is a combination—apply the max-cut theorem, delete the edges inside the smaller side so that it becomes independent, and feed the result to an independent-set completion bound. The completion bound itself is Razborov's Section 4.5 with a single step changed; Section 2 states precisely what is his and what is not. All scalar constants are rational and exactly certified. This is not a solution of Erdős problem 128, and no claim of priority is made.

1 Statement and normalization

For $n > 0$ let

$$\beta_*(G) = \min\Big\{n^{-2} \sum_{uv\in E(G)} z_u z_v \ :\ 0 \le z_v \le 1,\ \sum_v z_v = n/2\Big\}.$$

Proposition 1 (rounding)**.** *Every graph G on $n > 0$ vertices has a set of exactly $\lfloor n/2\rfloor$ vertices spanning at most $\beta_*(G)n^2$ edges.*

Proof. Fix a minimizer and two coordinates $z_u, z_v \in (0, 1)$. Transferring s from one to the other changes the objective by an affine function of s when $uv \notin E$, and by one with s^2-coefficient -1 when $uv \in E$; in both cases the cost is concave, so some endpoint of the feasible range of s is at least as good, and there z_u or z_v leaves $(0, 1)$. Iterating gives a minimizer with at most one coordinate in $(0, 1)$. If n is even that coordinate must be 0 and the minimizer is integral. If n is odd it equals $1/2$; deleting it removes only non-negative terms and leaves exactly $\lfloor n/2\rfloor$ selected vertices. □

Proposition 2 (blow-ups)**.** *Let $G[t]$ be the balanced t-fold blow-up. Then $\beta_*(G[t]) = \beta_*(G)$ and $D_2(G[t]) = t^2 D_2(G)$, where D_2 is the minimum number of edges whose deletion makes the graph bipartite.*

Proof. Each twin class is independent, so the objective on $G[t]$ depends on a profile only through the class sums $tw_u = \sum_{i\in u} z_i$; averaging inside classes leaves it unchanged and gives a profile on G of the same normalized cost, and lifting is the converse. For D_2, a bipartition of $G[t]$ splitting class u into a_u and $t-a_u$ has internal cost $\sum_{uv\in E(G)} \big(a_u a_v + (t-a_u)(t-a_v)\big)$, which is affine in each a_u separately; hence a minimum is attained with every $a_u \in \{0, t\}$, i.e. at a lifted bipartition of G, of cost $t^2 D_2(G)$. $\square$

Theorem 1. *Assume the Balogh–Clemen–Lidický theorem [**?**, Theorem 2(a)]: for n large enough, every triangle-free graph satisfies $D_2(G) \le n^2/23.5$. Then every finite triangle-free graph G has a set of $\lfloor n/2\rfloor$ vertices spanning at most $2587n^2/100000$ edges, strictly so for $n > 0$.*

For $n = 0$ the empty set gives the non-strict statement. The target $n^2/50$ remains unproved; so does any bound below $2587/100000$.

Remark 1. By Proposition **??** the hypothesis "n large enough" in [**?**] costs nothing: given any triangle-free G of order n, apply the cited bound to $G[t]$ for t large and divide by t^2 to get $D_2(G) \le 2n^2/47$. We use $I = 2/47$ throughout. Similarly, Proposition **??** lets us work with the relaxation β_* and only round at the very end.

2 The completion envelope, and what is new in it

For $\alpha \in [1/3, 1/2]$ put

$$g_\alpha(p) = 2\big(\tfrac12 - \alpha\big)(p-\alpha) + p\big(\tfrac12 - p\big), \qquad f(\alpha) = \tfrac12 \max_{p\in[\alpha,1/2]} g_\alpha(p),$$

and $f(\alpha) = 0$ for $\alpha \ge 1/2$. Since g_α is concave with unconstrained maximum at $p = 3/4 - \alpha$, and $3/4 - \alpha \le \alpha$ exactly when $\alpha \ge 3/8$,

$$f(\alpha) = \begin{cases} \frac32\alpha^2 - \frac54\alpha + \frac{9}{32}, & \frac13 \le \alpha \le \frac38 \quad \text{(interior maximum at } p = \frac34 - \alpha\text{)},\\ \frac{\alpha}{2}\big(\frac12 - \alpha\big), & \frac38 \le \alpha \le \frac12 \quad \text{(maximum at the endpoint } p = \alpha\text{)}.\end{cases} \tag{1}$$

f is continuous at $3/8$, where both branches equal $3/128$, and decreasing on $[1/3, 1/2]$, with $f(1/3) = 1/32$ and $f(2/5) = 1/50$. Also $f(\alpha) - \frac{\alpha}{2}(\frac12 - \alpha) = 2(\alpha - \frac38)^2 \ge 0$, so $\frac{\alpha}{2}(\frac12 - \alpha) \le f(\alpha)$ on the whole interval.

Lemma 1. *Let G be triangle-free with normalized independence number $\alpha = \alpha(G)$. If $\alpha \geq 1/3$ then $\beta_*(G) \leq f(\alpha)$. Consequently $\beta_*(G) \leq f(x)$ whenever G has an independent set of size at least xn with $1/3 \leq x \leq 1/2$.*

Attribution. For $\alpha \geq 3/8$ this is exactly Razborov [**?**, Theorem 3.6], whose proof occupies his Section 4.5. Our statement differs from his only in the range $\alpha \in [1/3, 3/8)$, and the proof differs from his in exactly one step:

- *Case 1 (the changed step).* Razborov derives the concave bound $2\beta_* \leq g_\alpha(p_b)$ and observes that its maximum sits at $p_b = 3/4 - \alpha$, "which is $\leq \alpha$ since we assumed $\alpha \geq 3/8$"; he therefore substitutes $p_b := \alpha$. For $\alpha < 3/8$ that substitution is unavailable, but the maximum is then interior to $[\alpha, 1/2]$ and one may simply evaluate g_α there. This is the whole of the extension, and it is a one-line observation on his own display.

- *Case 2 (unchanged).* His Case 2 uses $\alpha \geq 3/8$ only in its final step, where the requirement is $\partial Q_1/\partial p_b|_{p_b=\alpha} = (1-3\alpha)/2 \leq 0$, i.e. $\alpha \geq 1/3$. It therefore applies verbatim on the wider range. We reproduce it below for self-containedness, not for novelty.

The genuinely new content of this note is not Lemma **??** but its combination with a max-cut theorem in Section 3.

Proof of Lemma **??**. The last sentence follows from the first because $\alpha \geq x$ and f is decreasing. If $\alpha \geq 1/2$ then $\beta_*(G) = 0$. By Proposition **??** we may assume n even. Densities: all vertex counts and degrees are divided by n, all edge counts by n^2, and each edge is counted once.

Let A be a maximum independent set, $k = \frac{1}{2} - \alpha$, and grow $B \supseteq A$ by repeatedly adding a vertex with at most k neighbours in the current set, stopping at size $\frac{1}{2}$ or when no such vertex exists. In the first case the cost is at most $k^2 \leq \alpha k/2 \leq f(\alpha)$, using $\alpha \geq 1/3$ for the middle inequality. Otherwise we obtain B with

$$b \in [\alpha, \tfrac{1}{2}), \qquad e(B) \leq k(b-\alpha), \qquad e_B(v) > k \ \text{ for every } v \notin B. \tag{2}$$

Vertices outside B with more than $b/2$ neighbours in B are pairwise non-adjacent—two of them share a neighbour in B—so they form an independent set C with $|C|/n \leq \alpha$. Hence we may pick D disjoint from $B \cup C$ with $d = 1 - \alpha - b$, and every $z \in D$ has $k < s_z \leq b/2$, where s_z is its normalized degree into B. Write $h = \frac{1}{2} - b > 0$ and note $d - h = k > 0$.

Case 1: some $v \in B$ has at least hn neighbours in D. Pick hn of them; they lie in $N(v)$, hence are independent, and lie in D, hence send at most $b/2$ each into B. Their union with B has size $b + h = \frac{1}{2}$ and cost at most

$$H = k(b - \alpha) + \tfrac{b}{2}h = \tfrac{1}{2}g_\alpha(b) \le f(\alpha),$$

the last step by the definition of f as a maximum over $p \in [\alpha, 1/2]$, which contains b.

Case 2: every $v \in B$ has fewer than hn neighbours in D. Fix $v \in B$, set $E = N(v) \cap D$ and $r = |E|/n < h$, and take the profile equal to 1 on $B \cup E$, to $w = (h-r)/(d-r) \in [0, 1]$ on $D \setminus E$, and to 0 elsewhere; it sums to $b + r + (h - r) = \frac{1}{2}$. Accounting for the six kinds of pair—inside B, B to E, inside E (empty, as $E \subseteq N(v)$ is independent), B to $D \setminus E$, E to $D \setminus E$, and inside $D \setminus E$ (Mantel)—the cost is at most

$$k(b - \alpha) + m(B, E) + \tfrac{b}{2}(h - r) + r(h - r) + \tfrac{(h-r)^2}{4}.$$

For Mantel: in a triangle-free graph of order $N > 0$ with m edges every edge has endpoint degree sum at most N, so $4m^2/N \le \sum_v \deg(v)^2 \le Nm$ and $m \le N^2/4$.

Now average over $v \in B$. The displayed tail is concave in r (its r^2-coefficient is $-3/4$), so it may be evaluated at $\bar{r}$. A vertex $z \in D$ enters E with probability s_z/b, whence $\bar{r} = \frac{1}{bn}\sum_{z \in D} s_z \in [0, d/2]$ and, using $s_z^2 \le (k + \frac{b}{2})s_z - \frac{kb}{2}$—which is exactly $(s_z - k)(s_z - \frac{b}{2}) \le 0$—

$$\mathbb{E}\, m(B, E) = \tfrac{1}{bn}\textstyle\sum_{z \in D} s_z^2 \le \big(k + \tfrac{b}{2}\big)\bar{r} - \tfrac{kd}{2}.$$

Collecting terms, some half costs at most

$$Q = k\big(b - \alpha - \tfrac{d}{2}\big) + \tfrac{bh}{2} + \tfrac{h^2}{4} + \big(k + \tfrac{h}{2}\big)\bar{r} - \tfrac{3}{4}\bar{r}^{\,2}.$$

Q is concave in $\bar{r}$ with $\partial Q/\partial \bar{r} = (b - \alpha)/4 \ge 0$ at $\bar{r} = d/2$, so it increases on $[0, d/2]$ and

$$Q \le \tfrac{13}{16}\alpha^2 - \tfrac{9}{8}\alpha b - \tfrac{3}{16}b^2 - \tfrac{1}{4}\alpha + \tfrac{1}{2}b = \tfrac{\alpha k}{2} + \tfrac{1-3\alpha}{2}(b - \alpha) - \tfrac{3}{16}(b - \alpha)^2.$$

Both correction terms are ≤ 0 because $\alpha \ge 1/3$ and $b \ge \alpha$; hence $Q \le \alpha k/2 \le f(\alpha)$. $\square$

3 Deleting one side of a cut

This section is the new step. Let $I = 2/47$ and take a bipartition X, Y with $|X| \le |Y|$ realizing $D_2(G)$, with internal edge densities $q = e(X)/n^2$ and $j = e(Y)/n^2$, so $q + j \le I$. Put

$$c = \frac{n}{2|Y|} \in [\tfrac{1}{2}, 1], \qquad x = \frac{|X|}{n} = 1 - \frac{1}{2c}.$$

The profile equal to c on Y and 0 elsewhere is feasible and gives

$$\beta_*(G) \le c^2 j. \tag{3}$$

Suppose $c \ge 3/4$, equivalently $x \ge 1/3$. Delete the edges inside X: the resulting graph is still triangle-free and now has X as an independent set of density $x \in [1/3, 1/2]$, so Lemma **??** applies to it, and restoring the deleted edges costs at most q because all weights are at most 1:

$$\beta_*(G) \le f(x) + q. \tag{4}$$

Adding (**??**) to c^2 times (**??**) and using $q + j \le I$,

$$\beta_*(G) \le \frac{c^2\big(I + f(x)\big)}{1 + c^2}, \qquad c \ge \tfrac{3}{4}. \tag{5}$$

Neither ingredient is new—[**?**] is cited in [**?**]—but we have not found this combination in the literature. Note that (**??**) alone is weak near $c = 1$ and (**??**) alone is weak near $c = 3/4$; the point is that the two are strong in complementary regimes, and (**??**) interpolates.

4 Exact scalar certificate

Put $T = 2587/100000$ and $c_0 = 7797/10000$. The three intervals below cover $[\frac{1}{2}, 1]$.

For $c \le c_0$, (**??**) and $j \le I$ give $\beta_* \le Ic^2 \le Ic_0^2$, and $T - Ic_0^2 = \frac{1291}{2350000000} > 0$.

For $c_0 \le c \le \frac{4}{5}$ we have $x \le 3/8$, so the first branch of (**??**) applies, and the numerator of $T - c^2(I + f(x))/(1 + c^2)$ is

$$P(c) = \Big(T - I - \tfrac{17}{32}\Big)c^2 + \tfrac{7c}{8} + T - \tfrac{3}{8},$$

concave, hence minimized at an endpoint; both are positive:

$$P(c_0) = \frac{312483613}{235000000000000}, \qquad P\big(\tfrac{4}{5}\big) = \frac{22649}{117500000}.$$

For $\frac{4}{5} \le c \le 1$ the second branch applies and the numerator is $Ac^2 - Bc + C$ with

$$A = T + \tfrac{1}{4} - I > 0, \quad B = \tfrac{3}{8}, \quad C = T + \tfrac{1}{8}, \quad 4AC - B^2 = \frac{442569}{2500000000} > 0,$$

so it is positive on all of $\mathbb{R}$. Proposition **??** converts the resulting bound on β_* into the statement of Theorem **??**.

Remark 2. T is not a free choice: the supremum over $c \in [\frac{1}{2}, 1]$ of the bound $\min\{Ic^2, c^2(I + f(x))/(1 + c^2)\}$ equals $0.0258691\ldots$, attained where the two branches cross near c_0. So 2587/100000 is this argument's own optimum, rounded up, and no better constant follows from it without a new ingredient.

5 Verification, scope and limitations

`outputs/verify-cut-independent.py` and the independently written `outputs/verify-independent` recompute the identities, the rational margins and finite instances; both use exact arithmetic only. `formal/Scalar.lean` carries a compiled Lean 4 certificate for the scalar optimization of Section 4 and for the polynomial identities of Section 2, conditional on scalar hypotheses supplied as arguments.

Not established here: the conjecture 1/50; any counterexample; a formal proof of the graph-theoretic content of Sections 2 and 3; and the Balogh–Clemen–Lidický theorem, which is assumed as published. The literature audit underlying the attribution in Section 2 is recorded in `LITERATURE.md`; it is not a substitute for review by a specialist.

Acknowledgement of method

This work was carried out with substantial assistance from AI coding and reasoning agents, which produced and checked the algebra, the rational certificates and the Lean development, and performed the literature audit recorded in `LITERATURE.md`. The author takes responsibility for the mathematical content. No claim of priority is made and the result has not yet been read by a specialist referee.

References

[1] J. Balogh, F. C. Clemen, B. Lidický, *Max cuts in triangle-free graphs*, arXiv:2103.14179, Theorem 2(a).

[2] A. A. Razborov, *More about sparse halves in triangle-free graphs*, arXiv:2104.09406, Theorem 3.6 and Section 4.5.